PRODUCT STRATEGY & ARCHITECTURE

EduMore Decisions

Product vision, locked decisions and target architecture

20	3	PHASED
ARCHITECTURE DECISIONS	PRODUCT WORKFLOWS	DELIVERY MODEL

Planning baseline • Version 1.1 • 20 October 2026

Executive Summary

EduMore is a multi-institution academic productivity platform for faculty and management. It is designed to turn research and academic planning into reviewed, versioned, publishable learning resources—while keeping reliability, source traceability, security, and human approval central.

01 Create	02 Review	03 Operate
Research-led curricula, structured notes, technical diagrams, and assessment packages.	Automated quality checks plus faculty approval before publication of notes, tests, and official reports.	Academic progress, workload, planning, reports, task tracking, and recovery-aware background jobs.

Product principles

- Quality-first delivery: do not expand scope at the expense of content accuracy, usability, security, or test coverage.
- Never claim the system can never fail. Design for graceful degradation, recovery, observability, and clear error states; no software can guarantee zero failures.
- Do not fabricate sources, citations, answers, status updates, or success messages. Mark uncertain or unverifiable content for review.
- Use original synthesis rather than copying source material verbatim; respect source attribution, licensing, and applicable terms.
- Require server-side authorization for every protected operation. Hiding UI elements is not a security boundary.
- All long-running jobs need persisted status, bounded retries, idempotent side effects, and a user-visible outcome.
- Published resources must be versioned. Never silently overwrite approved material.
- Treat model outputs and web content as untrusted input; validate, sanitize, and constrain them before storage, rendering, or execution.

Decision status

Decisions 1–19: selected in the role-play. Decision 20: Notes & PDF Generator is the recommended first workflow, but the user had not explicitly confirmed that final choice when these documents were generated.

The 20 Architecture Decisions

This is the consolidated decision record. The descriptions below capture the selected direction; implementation details and vendors remain open until technical design.

01	**1. Curriculum research workflow** Research relevant online sources, check relevance and credibility, generate a structured curriculum and visual roadmap, then send it for faculty review.
02	**2. Notes and PDF workflow** Use a research-first workflow: collect sources, synthesize original notes, validate content, request faculty approval, and generate the final PDF only after approval.
03	**3. Diagram generation** Use a hybrid approach: structured code-based diagrams for technical material and AI-generated illustrations where they add value. Validate readability and technical consistency.
04	**4. Assessment generation** Generate a complete assessment package: topic coverage, difficulty levels, questions, answer key, explanations, marks, time limits, duplicate checks, faculty approval, and downloadable PDF.
05	**5. Academic management** Support syllabus and course progress, faculty workload, academic reports, timetable/course planning, and delay or pending-work alerts.
06	**6. Approval policy** Use smart approval: drafts and internal suggestions may be generated automatically; notes, tests, and official reports require faculty approval before publication.
07	**7. AI workflow architecture** Use separate specialized workflows for curriculum, notes/assessments, and academic management. Reuse shared research, validation, PDF, storage, and observability services.
08	**8. Data storage** Use PostgreSQL for structured records and object storage for PDFs and generated files. Add semantic/vector search only when a demonstrated requirement justifies it.
09	**9. Research and sources** Use multi-source research with trusted-source prioritization, cross-source verification, citations, conflict detection, and explicit uncertainty flags.
10	**10. Quality validation** Combine deterministic rules, AI review, source checks, bounded revision attempts, and faculty approval. Flag unresolved problems instead of retrying forever.

11	**11. Multi-institution access** Support Admin, Management, and Faculty roles across multiple institutions. Enforce tenant isolation in backend access and data queries.
12	**12. Execution strategy** Use synchronous APIs for quick operations and background jobs for long-running research, content generation, assessment generation, and PDF creation. Expose job status.
13	**13. Model strategy** Route tasks to appropriate models based on quality, complexity, latency, and cost. Support controlled provider fallback and validate fallback outputs.
14	**14. Workflow recovery** Persist checkpoints for long-running workflows and resume from the last safe checkpoint. Use retry limits, idempotency, task state, and safe failure handling.
15	**15. Resource versioning** Link learning resources to institution, course, curriculum version, and chapter. Preserve history and flag potentially outdated resources when a curriculum changes.
16	**16. Observability and audit** Track operational metrics, errors, latency, model usage/cost, approvals, content revisions, and access events with privacy-safe institution-aware audit logs.
17	**17. Security** Use layered security: authentication, password hashing, role permissions, tenant isolation, secure file access, encryption, secrets management, rate limiting, and security tests.
18	**18. Dashboard and task center** Provide role-based dashboards plus a task center showing pending approvals, running/failed jobs, alerts, and permitted actions such as retry, review, approve, or download.
19	**19. Phased release** Release in phases with acceptance gates. Start with a complete end-to-end workflow, then add remaining AI workflows and academic management after tests and quality gates pass.
20	**20. First end-to-end workflow** Selected by the architect recommendation for this planning handoff: Notes & PDF Generator first, because it exercises research, diagrams, validation, approval, background processing, storage, and PDF generation. This choice was recommended but not separately confirmed by the user.

Target Architecture

LAYER	RESPONSIBILITY
React frontend	Role-based dashboards, task center, editors, approval screens, progress and downloads.
FastAPI application	Authentication, authorization, institution context, APIs, validation, audit events and job submission.
Specialized workflows	LangGraph workflows for curriculum, notes/assessment, and academic management where orchestration is needed.
Shared services	Research and source verification, model routing, quality checks, diagrams, PDFs, notifications and audit.
Background execution	Queue/workers for long-running jobs, persisted status, checkpoint recovery, bounded retries and idempotency.
Persistence	PostgreSQL for structured data; object storage for generated files. Vector search deferred until justified.
Operations & security	Monitoring, privacy-safe logs, provider-cost tracking, secrets management, encryption, tenant isolation and tests.

Phased delivery roadmap

01	**Foundation**	Authentication, tenant isolation, data model, jobs, recovery, security, monitoring, versioning.
02	**First end-to-end workflow**	Research → notes → diagrams → quality review → faculty approval → PDF.
03	**Remaining AI workflows**	Curriculum Designer and complete assessment/test generation.
04	**Academic Management**	Progress, workload, scheduling, alerts, and management reports.

Reliability is an engineering target, not an absolute promise

No software can honestly guarantee zero failures. EduMore must prevent avoidable failures, detect incidents, preserve valid work, recover safely where possible, and never claim success when a required stage failed.

Open technical decisions

- Choose hosting, database and object-storage providers.
- Choose the queue/worker technology and checkpoint persistence implementation.
- Select model providers and define evaluation sets, cost limits and fallback policy.
- Define availability, latency, backup/restore, retention and incident-response objectives.
- Confirm privacy, licensing and institution-specific compliance requirements.